최적제어 숙제3 이윤경 2021321216 제출마감일: 21.10.15

$\boxed{4.2}$ prove the fundamental lemma; that is, show that if $h(t)$ is continuous for $t \in (t_0 \sim t_f)$, and if $\int_{t_0}^{t_f} h(t)\delta x(t)\,dt = 0$ for every function $\delta x(t)$ that is continuous in $t_0 \sim t_f$ with $\delta x(t_0) = \delta x(t_f) = 0$, then $h(t)$ must be identically zero in $t_0 \sim t_f$

$h(t) \neq 0$ 이라고 가정하자. t는 $\underline{t_1 \sim t_2} \subset \underline{t_0 \sim t_f}$ 이다. then $\int_{t_0}^{t_f} h(t)\,\delta x(t)\,dt = \int_{t_1}^{t_2} h(t)\,\delta x(t)\,dt$ 이다.

이때, 우변은 0이 아닐수도 있는게, $h(t) \neq 0$ ($t_1 \sim t_2$에서) 이기 때문이다. 이렇게 되면 좌변도 0이 아닐수 있게되므로, 문제에서 주어진 좌변식$=0$ 이라함과 모순된다. ∴ $h(t) \neq 0$ 이 아니어야 하므로 $h(t) = 0$ 이다. $t_0 \sim t_f$ 동안.

$\boxed{4.5}$ consider problem 1 of section 4.2 and let n be a specified continuously differentiable function that is arbitrary in $t_0 \sim t_f$, except at end points. where $n(t_0) = n(t_f) = 0$. If e is arbitrary real parameter, then $x^* + en$ represents a family of curves. Evaluating the functional $J(x) = \int_{t_0}^{t_f} g(x, \dot{x}, t)\,dt$ on the family $x^* + en$ makes J a function of e, and if x^* is extremal, this function must have a relative extremum at point $e=0$. Show that Euler eq (4.2.10) is obtained from $N\left(\boxed{\left.\dfrac{dJ(x^*+en)}{de}\right|_{e=0} = 0}\right.$

eq.4.2.10 $\rightarrow \dfrac{\partial g}{\partial x}\left(x^*(t), \dot{x}^*(t), t\right) - \dfrac{d}{dt}\left(\dfrac{\partial g}{\partial \dot{x}}\left(x^*(t), \dot{x}^*(t), t\right)\right) = 0.$

$$J(x^* + en) = \int_{t_0}^{t_f} g\left(x^*(t) + en(t),\ \dot{x}^*(t) + e\dot{n}(t),\ t\right) dt. \quad , \quad \frac{dJ(x^*+en)}{de} = \frac{d}{de}\int_{t_0}^{t_f} g\left(x^*(t) + en(t),\ \dot{x}^*(t) + e\dot{n}(t),\ t\right) dt. \Longrightarrow$$

$$\Longrightarrow = \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\left(x^*(t) + en(t),\ \dot{x}^*(t) + e\dot{n}(t),\ t\right) + \frac{\partial g}{\partial \dot{x}}\left(x^*(t) + en(t),\ \dot{x}^*(t) + e\dot{n}(t),\ t\right)\dot{n}(t)\right] dt.$$

$$\left.\frac{dJ(x^*+en)}{de}\right|_{e=0} = \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}\left(x^*(t), \dot{x}^*(t), t\right) n(t) + \frac{\partial g}{\partial \dot{x}}\left(x^*(t), \dot{x}^*(t), t\right)\dot{n}(t)\right] dt = 0.$$

$\boxed{\text{부분적분으로, } u = \dfrac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t)\ ,\ \dot{v} = \dot{n}.}$ 설정.

$$\int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}(x^*, \dot{x}^*, t)\,n + \frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t)\,\dot{n}\right] dt = \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}(x^*, \dot{x}^*, t)\,n\right] dt + \left.\frac{\partial g}{\partial \dot{x}}\,n(t)\right|_{t=t_0}^{t=t_f} - \int_{t_0}^{t_f}\left[\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t)\right) n(t)\right] dt$$

$$= \int_{t_0}^{t_f}\left[\frac{\partial g}{\partial x}(x^*, \dot{x}^*, t)\,n - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t)\right) n\right] dt = \int_{t_0}^{t_f}\left[\underbrace{\frac{\partial g}{\partial x}(x^*, \dot{x}^*, t) - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t)\right)}_{=\,0\ (\text{오일러 식이네})}\right] \underbrace{n}_{\text{임의}}\, dt = 0.$$

ex4.3.3 Find extremal curve for $J=\int_0^{\frac{\pi}{4}}\left(x_1^2+\dot{x}_1\dot{x}_2+\dot{x}_2^2\right)dt$ $\quad x_1(0)=1$, $x_2(0)=1.5$, ~~$x_1(\frac{\pi}{4})=2$~~, $x_2(\frac{\pi}{4})=$ 자유

case 2 변형 판 (~~원래는 x_1 관계만 소관계~~) ∵ x가 x_1/x_2로 여러개로 넥 터리서

① 오일러식 → $\frac{\partial g}{\partial x_1}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}_1}\right)=0=2x_1-\frac{d}{dt}\dot{x}_2=2x_1-\ddot{x}_2=0$

② 오일러식 → $\frac{\partial g}{\partial x_2}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}_2}\right)=0=0-\frac{d}{dt}\left(\dot{x}_1+2\dot{x}_2\right)=-\ddot{x}_1-2\ddot{x}_2=0$

$\ddot{x}_2=-\frac{\ddot{x}_1}{2}$

$2x_1=\ddot{x}_2$, $-\ddot{x}_1=2\ddot{x}_2$ → $2x_1=-0.5\ddot{x}_1$ → $-4x_1=\ddot{x}_1$

$\boxed{\frac{\partial g}{\partial \dot{x}_2}=0}$

$\boxed{x_1=C_1\cos 2t+C_2\sin 2t}$

$\dot{x}_1=-2C_1\sin 2t+2C_2\cos 2t$

$\ddot{x}_1=-4C_1\cos 2t-4C_2\sin 2t$

$\ddot{x}_2=2C_1\cos 2t+2C_2\sin 2t$

$\dot{x}_2=C_1\sin 2t-C_2\cos 2t+C_3$

$\boxed{x_2=-\frac{C_1}{2}\cos 2t-\frac{C_2}{2}\sin 2t+C_3t+C_4}$

식4.3.18 → $$\boxed{\delta J=\left[\frac{\partial g}{\partial \dot{x}_1},\ \frac{\partial g}{\partial \dot{x}_2}\right]\begin{bmatrix}\delta x_1(t_f)\\ \delta x_2(t_f)\end{bmatrix}+\left(g-\left[\frac{\partial g}{\partial \dot{x}_1},\ \frac{\partial g}{\partial \dot{x}_2}\right]\begin{bmatrix}\dot{x}_1(t_f)\\ \dot{x}_2(t_f)\end{bmatrix}\right)\delta t_f=0.}$$

이때, t_f와 $x_1(t_f)$=고정되므로, $\delta t_f=0$, $\delta x_1(t_f)=0$ 이다.

$\therefore \delta J=\underbrace{\frac{\partial g}{\partial \dot{x}_2}}_{0}\cdot\underbrace{\delta x_2(t_f)}_{\text{임의}}=0.$ [∵ $\delta x_2(t_f)$=임의], $\therefore \frac{\partial g}{\partial \dot{x}_2}=0$ 이는 횡단식인 $\left(\frac{\partial g}{\partial \dot{x}}=0\right)$과 유사하다.

i) $x_1(0)=1$, $x_1(\frac{\pi}{4})=2$ ⟶ $x_1=C_1\cos 2t+C_2\sin 2t$ → $x_1(0)=1=C_1$, $x_1(\frac{\pi}{4})=2=C_2$

ii) $x_2(0)=1.5$, $\frac{\partial g}{\partial \dot{x}_2}=0$ ⟶ $x_2=-\frac{C_1}{2}\cos 2t-\frac{C_2}{2}\sin 2t+C_3t+C_4=-0.5\cos 2t-\sin 2t+C_3t+C_4$ ⟹ $x_2(0)=1.5=C_4-0.5$, $C_4=2$

$\frac{\partial}{\partial \dot{x}_2}\left(x_1^2+\dot{x}_1\dot{x}_2+\dot{x}_2^2\right)=\dot{x}_1+2\dot{x}_2=0$ (at $t=t_f$)

$=-2C_1\sin 2t+2C_2\cos 2t+2C_1\sin 2t-2C_2\cos 2t+2C_3$

$=-2\sin 2t+4\cos 2t+2\sin 2t-4\cos 2t+2C_3$

$=-2+0+2+0+2C_3=0$, $\therefore C_3=0$

$\therefore x_1=\cos 2t+2\sin 2t$

$\therefore x_2=-\frac{1}{2}\cos 2t-\sin 2t+2$

ex 4.4.1 Find piecewise smooth curve that begins at point $x(0)=0$, ends at point $x(2)=1$, and minimizes $J=\int_0^2\dot{x}^2(1-\dot{x})^2\,dt$.

→ 4가지 경우 (불연속)

오일러식 → $\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)=0=0-\frac{d}{dt}\left(2\dot{x}-6\dot{x}^2+4\dot{x}^3\right)=-2\ddot{x}+12\dot{x}\ddot{x}-12\dot{x}^2\ddot{x}=\ddot{x}\underbrace{\left(-2+12\dot{x}-12\dot{x}^2\right)}_{\text{실근}}$ ⟶ $\ddot{x}=0$

$\dot{x}=C_1$

$\boxed{x=C_1t+C_2}$

$C_2=0$

$C_1=0.5$

$g=\dot{x}^2(1-\dot{x})^2=\dot{x}^2(1-2\dot{x}+\dot{x}^2)=\dot{x}^2-2\dot{x}^3+\dot{x}^4$

WE식 → ① $\left.\frac{\partial g}{\partial \dot{x}}\right|_{t_1^+}=\left.\frac{\partial g}{\partial \dot{x}}\right|_{t_1^-}$

② $\left.g-\frac{\partial g}{\partial \dot{x}}\dot{x}\right|_{t_1^+}=\left.g-\frac{\partial g}{\partial \dot{x}}\dot{x}\right|_{t_1^-}$

→ $\frac{\partial g}{\partial \dot{x}}=2\dot{x}-6\dot{x}^2+4\dot{x}^3$

$g-\frac{\partial g}{\partial \dot{x}}\dot{x}=\dot{x}^2-2\dot{x}^3+\dot{x}^4-2\dot{x}\dot{x}+6\dot{x}^3-4\dot{x}^4$ ~~$=5\dot{x}^4-8\dot{x}^3+\dot{x}^2-2\dot{x}\dot{x}$~~

$=-3\dot{x}^4+4\dot{x}^3-\dot{x}^2=-3\dot{x}^4+4\dot{x}^3-\dot{x}^2$

BC 조건. $x(0)=0$, $x(2)=1$

① $\left.\frac{\partial g}{\partial \dot{x}}\right|_{t_1^+}=\left.\frac{\partial g}{\partial \dot{x}}\right|_{t_1^-}$ ⟹ $4\dot{x}^3(t_1^+)-6\dot{x}^2(t_1^+)+2\dot{x}(t_1^+)=4\dot{x}^3(t_1^-)-6\dot{x}^2(t_1^-)+2\dot{x}(t_1^-)$

$\dot{x}(4\dot{x}^2-6\dot{x}+2)=\dot{x}(4\dot{x}^2-6\dot{x}+2)$

$\dot{x}(2\dot{x}-1)(2\dot{x}-2)$ $\quad$ $\dot{x}(2\dot{x}-1)(2\dot{x}-2)$ ⟶ ∴ $\dot{x}(t_1^+)=0,\frac{1}{2},1$; $\dot{x}(t_1^-)=0,\frac{1}{2},1$

② $\left.g-\frac{\partial g}{\partial \dot{x}}\dot{x}\right|_{t_1^+}=\left.g-\frac{\partial g}{\partial \dot{x}}\dot{x}\right|_{t_1^-}$ $=-3\dot{x}^4(t_1^+)+4\dot{x}^3(t_1^+)-\dot{x}^2(t_1^+)=-3\dot{x}^4(t_1^-)+4\dot{x}^3(t_1^-)-\dot{x}^2(t_1^-)$

$-\dot{x}^2(3\dot{x}^2-4\dot{x}+1)=-\dot{x}^2(3\dot{x}-1)(\dot{x}-1)$ → ∴ $\dot{x}(t_1^+)=0,\frac{1}{3},1$; $\dot{x}(t_1^-)=0,\frac{1}{3},1$

교집합으로 0, 1 이 정답! 단, 동시에 같은 숫자는 겹쳐서 안됨.

∴ $\dot{x}(t_1^+)=0$, $\dot{x}(t_1^-)=1$ or $\dot{x}(t_1^+)=1$, $\dot{x}(t_1^-)=0$

$x(t_1^+)$=상수, $x(t_1^-)=t+$상수

then $t_1=1$: $1=C=x(4)$; $t+0=t+C=x(3t)$

$x(4)=t+C=t$; $x(3t)=C=0$

$x=\sin t$

ex 4.5.1 제한조건 Find point on line $y_1+y_2=5$ that is nearest the origin. (0,0)

점 제한

augmented function, the distance from origin squared, as. $f_a(y_1, y_2, p) = y_1^2 + y_2^2 + p(y_1+y_2-5)$

목표는 f_a의 extreme, and 그때의 y_1과 y_2이다. 제한조건이 만족되면 p항은 항상 0이다.

$\boxed{\therefore f_a = y_1^2 + y_2^2}$

$$df_a(y_1^*, y_2^*, p) = \underbrace{(2y_1^*+p)}_{\text{편미분}}\Delta y_1 + \underbrace{(2y_2^*+p)}_{\text{편미분}}\Delta y_2 + (y_1^*+y_2^*-5)\Delta p = 0.$$

이제 $2y_1^*+p=0$ or $2y_2^*+p=0$ 중에 아무거나 선택한다. (제한조건 만족되어 p항은 항상 0이다.)

$df_a(y_1^*, y_2^*, p) = (2y_1^*+p)\Delta y_1 = 0. \rightarrow \Delta y_1 \neq 0$ 이므로, $2y_1^*+p=0$ 이어야 한다. 즉, $\boxed{2y_1^*+p = 2y_2^*+p = 0}$

거리 $= \sqrt{2.5^2+2.5^2} = \frac{5}{\sqrt{2}}$ // $\leftarrow$ $y_1 = y_2 = 2.5$ $\leftarrow$ $y_1^*+y_2^*=5$ $\leftarrow$ $\therefore y_1^* = y_2^*$

---

ex 4.5.2 Find point in 3차원 유클리드 공간에서(원점근처에 있는) and lies on the intersection of surfaces $y_3 = y_1y_2+5$, $y_1+y_2+y_3=1$

점 제한

augmented function, the distance from origin squared, as. $f_a(y_1, y_2, y_3, p_1, p_2) = y_1^2+y_2^2+y_3^2 + p_1(y_1y_2-y_3+5) + p_2(y_1+y_2+y_3-1)$

목표는 f_a의 extreme, and 그때의 y_1과 y_2이다. 제한조건이 만족되면 p항은 항상 0이다.

$\boxed{\therefore f_a = y_1^2+y_2^2+y_3^2}$

$$df_a(y_1^*, y_2^*, y_3^*, p_1, p_2) = \underbrace{(2y_1+p_1y_2+p_2)}_{0\ ①}\Delta y_1 + \underbrace{(2y_2+p_1y_1+p_2)}_{0\ ②}\Delta y_2 + \underbrace{(2y_3-p_1+p_2)}_{0\ ③}\Delta y_3 + \underbrace{(y_1y_2-y_3+5)}_{0\ ④}\Delta p_1 + \underbrace{(y_1+y_2+y_3-1)}_{0\ ⑤}\Delta p_2 = 0.$$

이제 $2y_1+p_1y_2+p_2=0$ or $2y_2+p_1y_1+p_2=0$ or $2y_3-p_1+p_2=0$ 중에 아무거나 선택한다. (제한조건 만족하여 p항은 항상 0이다.)

연립) ① = ② $\rightarrow 2y_1+p_1y_2+\cancel{p_2} = 2y_2+p_1y_1+\cancel{p_2}=0$

$(2-p_1)y_1+\cancel{p_2} = (2-p_1)y_2+\cancel{p_2} \rightarrow \boxed{y_1=y_2}$ or $\boxed{p_1=2}$
X O.

→ ④, ⑤식에 대입
i) $y^2+5=y_3=1-2y \rightarrow y^2+2y+4=0$, y = 허수.
즉, $y_1 \neq y_2$ $\therefore p_1=2$

조건
$2y_1+2y_2+p_2=0 \rightarrow 2y_1+2y_2+p_2=0$
~~$2y_2+2y_1+p_2=0$~~
$2y_3+p_2-2=0 \rightarrow ③$
$y_1y_2-y_3+5=0$
$y_1+y_2+y_3-1=0 \rightarrow 2y_1+2y_2+2y_3-2=0$

$\rightarrow p_2-2y_3+2=0 \rightarrow 2p_2=0$, $p_2=0$, $y_3=1$
③ $p_2+2y_3-2=0$

$y_1y_2=-4$
$y_1+y_2=0$

$y_1=2, y_2=-2$
or
$y_1=-2, y_2=2$

점 제한

거리 $= \sqrt{2^2+2^2+1} = \sqrt{9} = 3$ $\leftarrow$ $\therefore (y_1, y_2, y_3) = (2,-2,1)$ or $(-2,2,1)$ //

---

ex 4.5.3 Find necessary condition that must be satisfied by curve of smallest length which lies on sphere. $w_1^2+w_2^2+t^2=R^2$, $(t = t_0 \sim t_f)$, and joins specified points w_0, t_0 and w_f, t_f.

Length curve is given below, and it is functional to be minimized.

$$J(w) = \int_{t_0}^{t_f} \sqrt{\left(\frac{dt}{dt}\right)^2 + \left(\frac{dw_1}{dt}\right)^2 + \left(\frac{dw_2}{dt}\right)^2}\, dt = \int_{t_0}^{t_f} \sqrt{1+\dot{w}_1^2+\dot{w}_2^2}\, dt$$

$$g_a(w, \dot{w}, p, t) = \sqrt{1+\dot{w}_1^2+\dot{w}_2^2} + p(w_1^2+w_2^2+t^2-R^2)$$

EL식

$\frac{\partial g_a}{\partial w_1} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{w}_1}\right) = 0 = 2pw_1 - \frac{d}{dt}\left(\frac{2\dot{w}_1}{2\sqrt{1+\dot{w}_1^2+\dot{w}_2^2}}\right) = 0$

$\frac{\partial g_a}{\partial w_2} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{w}_2}\right) = 0 = 2pw_2 - \frac{d}{dt}\left(\frac{2\dot{w}_2}{2\sqrt{1+\dot{w}_1^2+\dot{w}_2^2}}\right) = 0$

$w_1^2+w_2^2+t^2=R^2$

이 3개는 $w_1^{2*}+w_2^{2*}+t^2=R^2$ 의 곡선을 이루는 NC조건이다.

ex4.5.5 $\dot{x}_1 = x_2 - x_1$ is controlled to minimized performance measure $J(x,u) = \int_{t_0}^{t_f} \frac{1}{2}(x_1^2 + x_2^2 + u^2)\,dt$.

미방제한 $\dot{x}_2 = -2x_1 - 3x_2 + u$ Find NC for optimal control.

$$g_a = \frac{1}{2}(x_1^2 + x_2^2 + u^2) + p_1(\dot{x}_1 - x_2 + x_1) + p_2(\dot{x}_2 + 2x_1 + 3x_2 - u)$$

EL식

$$\frac{\partial g_a}{\partial x_1} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{x}_1}\right) = x_1 + p_1 + 2p_2 - \frac{d}{dt}(p_1) = x_1 + p_1 + 2p_2 - \dot{p}_1 = 0$$

$$\frac{\partial g_a}{\partial x_2} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{x}_2}\right) = x_2 - p_1 + 3p_2 - \frac{d}{dt}(p_2) = x_2 - p_1 + 3p_2 - \dot{p}_2 = 0$$

$$\frac{\partial g_a}{\partial u} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{u}}\right) = u - p_2 - 0 = 0$$

3개와 더불어 → $\boxed{\begin{array}{l} \dot{x}_1 - x_2 + x_1 = 0 \\ \dot{x}_2 + 2x_1 + 3x_2 - u = 0 \end{array}}$

총 5개가 NC이다.

ex4.5.17 $\dot{x}_1 = -x_1 + x_2 + u$ is controlled to minimize performance measure $J(x,u) = \int_{t_0}^{t_f} \frac{1}{2}(x_1^2 + x_2^2)\,dt$,

등주제한 $\dot{x}_2 = -2x_1 - 3x_2 + u$ Total control energy expended is $\int_{t_0}^{t_f} u^2\,dt = C$ = 상수. $\boxed{\text{Find NC for 최적제어}}$.

$x_1 = w_1$, $x_2 = w_2$, $u = w_3$ 로 설정. → $J(w) = \int_{t_0}^{t_f} \frac{1}{2}(w_1^2 + w_2^2)\,dt$.

→ $\dot{w}_1 = -w_1 + w_2 + w_3$ ④

→ $\dot{w}_2 = -2w_1 - 3w_2 + w_3$ ⑤, $\int_{t_0}^{t_f} w_3^2\,dt = C$.

$$g_a = \frac{1}{2}w_1^2 + \frac{1}{2}w_2^2 + p_1(-w_1 + w_2 + w_3 - \dot{w}_1) + p_2(-2w_1 - 3w_2 + w_3 - \dot{w}_2) + p_3(w_3^2 - \dot{z}(t))$$

④ ⑤ ⑥

$\dot{p}_3 = 0$ ⑦

미항의 미분=0 ∴ f=0

EL식

$$\frac{\partial g_a}{\partial w_1} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{w}_1}\right) = w_1 - p_1 - 2p_2 - \frac{d}{dt}(-p_1) = w_1 - p_1 - 2p_2 + \dot{p}_1 = 0 \quad ①$$

$$\frac{\partial g_a}{\partial w_2} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{w}_2}\right) = w_2 + p_1 - 3p_2 - \frac{d}{dt}(-p_2) = w_2 + p_1 - 3p_2 + \dot{p}_2 = w_2 + p_1 - 3p_2 + \dot{p}_2 = 0 \quad ②$$

$$\frac{\partial g_a}{\partial w_3} - \frac{d}{dt}\left(\frac{\partial g_a}{\partial \dot{w}_3}\right) = p_1 + p_2 + 2p_3 w_3 = 0 \quad ③$$

$z(t_0) = 0$ ⑧

$z(t_f) = C$ ⑨

B.C(조건)